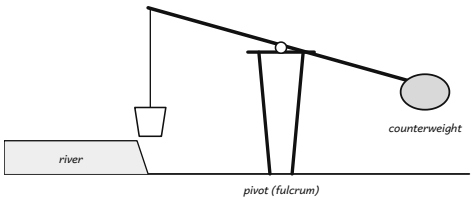
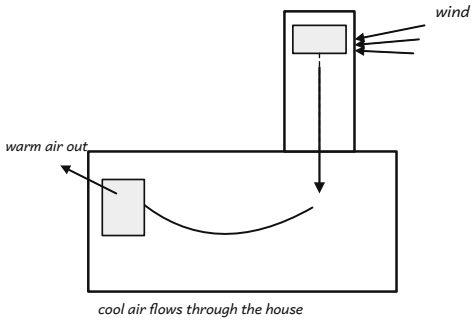
EARTH AND SPACE

Systems, Models and Technologies

Systems and models, their limits, and technologies from past civilisations that shape our lives today





In this pack

Learn it: Systems, models and clever technologies from the past ★☆☆
Fact check: Inventions and where they came from ★☆☆
Think it through: How did it work? Why does it matter? ★★☆
Investigate: Does a windcatcher model cool a room? ★★☆
Real-world case study: Bringing back Dhaka muslin ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Systems, models and clever technologies from the past

★☆☆ Grade 5

Name: Class: Date:

Learning goals

- Explain what a system and a model are, and the limits of models.
- Describe technologies from different civilisations.
- Explain how past technologies still affect our lives today.

A bicycle, a beehive, the water cycle and the human body are all **systems**: sets of parts that work together and affect each other. Scientists and engineers use **models** to understand systems and to predict what will happen.

Systems

Every system has **parts**, **connections** between them, and often **inputs** and **outputs**. In a tube-well, the input is your push on the handle, the parts are the handle, rod and valve, and the output is water. If one part breaks, the whole system may stop working.

Models

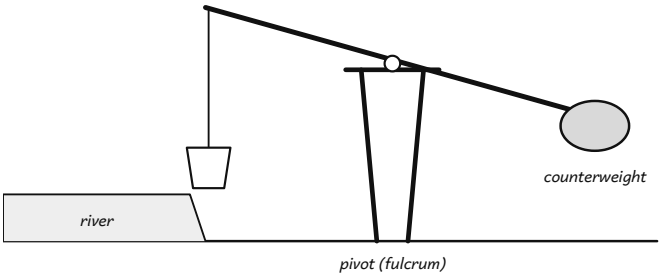
A **model** is a simpler way of showing something real. A **globe** models the Earth; a **map** models a town; a diagram of the water cycle models how water moves; weather forecasters use **computer models** to predict cyclones.

Think like a scientist

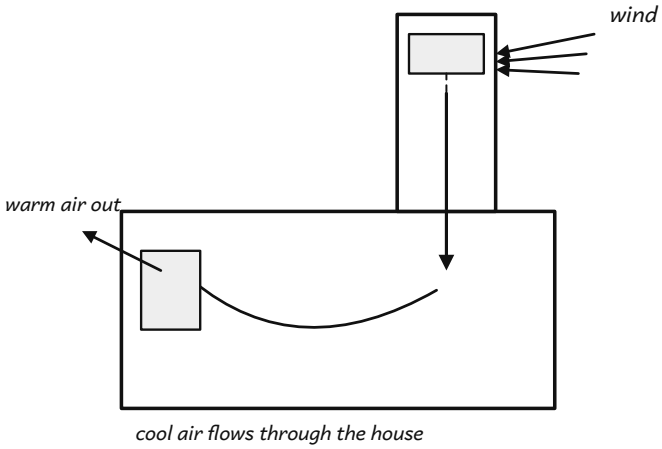
Every model has limits. A globe shows the shape of the Earth but not its real size or the clouds moving over it. Good scientists always ask: "What does this model show well? What does it leave out?"

Technologies from past civilisations

- **Indus Valley (about 4500 years ago):** cities such as Mohenjo-daro had planned streets, baked-brick houses and covered **drains** to carry dirty water away.
- **Ancient Egypt:** the **shadoof**, a long pole on a pivot with a bucket on one end and a heavy weight on the other, lifted water from the Nile to the fields. It is a **lever**.
- **Persia (Iran):** tall **windcatchers** caught the breeze and sent it down into houses to cool them, without electricity. Underground channels called **qanats** carried water across deserts.
- **China:** paper, the magnetic **compass** and printing.
- **The Islamic world (around 1200):** the engineer **al-Jazari** designed water-raising machines and clocks using gears and cranks.
- **Bengal:** for centuries weavers made **muslin**, a cotton cloth so fine that a whole sari could pass through a ring. Ancient cities such as **Wari-Bateshwar** traded beads across Asia more than 2000 years ago.



A shadoof: a lever for lifting water.



A windcatcher sends cool air through a house.

From the past to the present

Today's **sewers** follow the same idea as Indus drains. **Cranes** use the lever and pulley ideas of the shadoof. Architects design new buildings with windcatcher-style towers to save energy. Old ideas often inspire new solutions.

Key words

system	parts that work together
input	what goes into a system
output	what comes out of a system
model	a simpler representation of something real
limitation	something a model cannot show
technology	tools and inventions people make to solve problems
civilisation	a large, organised society

FACT CHECK · B6 · SYSTEMS AND MODELS

Inventions and where they came from

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Link technologies to civilisations.
- Identify parts of a system.

1 Match each technology to the civilisation it came from. [5]

Draw a line to match each pair, or write the letter.

covered drains in planned cities ●	● **A** Persia
shadoof ●	● **B** China
windcatchers and qanats ●	● **C** Bengal
magnetic compass ●	● **D** Ancient Egypt
fine muslin cloth ●	● **E** Indus Valley

2 Complete the table for a hand tube-well system. [3]

Inputs	Parts	Outputs

3 True or false? [4]

Statement	True	False	If false, correct it
1. A globe is a model of the Earth.			
2. Models always show everything about the real thing.			
3. A shadoof is a kind of lever.			
4. Windcatchers need electricity to work.			

4 Name three models you have used in school. [3]

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How did it work? Why does it matter?

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Explain how old technologies work.
- Evaluate models.

1 **Explain how a shadoof makes lifting water easier.** [2]

2 **Why were covered drains important for the health of people in Mohenjo-daro?** [2]

3 **Choose one model and complete the table.** [3]

Model	What it shows well	What it does not show

4 **A weather forecaster says, "Our computer model predicts the cyclone will reach the coast on Tuesday." Why do forecasts sometimes change as the storm gets closer?** [2]

5 **How could an old idea help solve a problem in Dhaka today? Give one example.** [2]

INVESTIGATE · B6 · SYSTEMS AND MODELS

Does a windcatcher model cool a room?

★★☆ Grade 5

Name: **Class:** **Date:**

Learning goals

- Test a model fairly.
- Interpret results.
- Evaluate the model.

Class 5B made three model houses from shoe boxes. They put each one 50 cm from a desk fan and a lamp for 20 minutes and then measured the temperature inside. All boxes started at 26 °C.

Model house	Temperature after 20 minutes (°C)
A: no tower, closed	34
B: windcatcher tower facing the fan	30
C: windcatcher tower with a wet cloth inside	27

1 **Draw a bar chart of the temperatures.** [4]



2 **Which house stayed coolest? Explain why, using the words *wind* and *evaporation*.** [3]

3 **Name two things the class kept the same.** [2]

4 **Give one limitation of this model compared with *a real* house.** [1]

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REAL-WORLD CASE STUDY · B6 · SYSTEMS AND MODELS

Bringing back Dhaka muslin

★★★
Grade 5

Name: Class: Date:

Learning goals

- Explain how a past technology connects to today.
- Consider history, science and economics together.
- Form an opinion with reasons.

Science | History | Design | Economics | Culture

For hundreds of years, weavers in the Dhaka region made **muslin**, one of the finest cloths in the world. It was woven from a special cotton plant called **phuti karpas**, which grew along the banks of the Meghna river. Spinners twisted its short fibres into threads so thin they were almost invisible, often working in the damp morning air so that the threads did not break.

Muslin was traded to kings and queens across Asia and Europe. But in the 1800s, cheap machine-made cloth from factories, changes in trade and the loss of the skilled workers caused muslin making to die out. Even the phuti karpas plant was thought to be lost.

Recently, Bangladeshi scientists and weavers searched for the plant, compared its DNA with old muslin samples in museums, grew it again and trained weavers. In 2021 they succeeded in weaving new Dhaka muslin.

Based on news reports about the Bangladesh muslin revival project.

1 **Draw a timeline of the story of Dhaka muslin with at least four events.** [3]

Timeline

2 **How did scientists use evidence to find the right cotton plant?** [2]

3 **Why did spinners work in damp air? Use what you know about materials.** [2]

4 **Muslin is very expensive and slow to make. Should the government keep supporting the revival? Give your opinion and two reasons.** [3]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

D	S	F	G	I	E	W	D	G	M	B
R	Y	S	J	M	O	D	E	L	U	K
A	S	K	B	C	F	N	C	Y	S	J
I	T	I	W	O	M	P	C	S	L	C
N	E	S	V	M	H	H	S	I	I	T
F	M	J	B	P	L	P	T	O	N	M
P	H	S	N	A	E	R	V	U	G	O
C	O	G	H	S	V	P	K	T	L	G
L	U	A	T	S	E	T	R	P	O	N
B	K	T	F	S	R	H	U	U	B	K
L	F	G	J	I	N	P	U	T	E	N

compass drain globe input lever model muslin output system

2 **Which is a model of the Earth?** [1]

- ◯ a compass
- ◯ a globe
- ◯ a shadoof
- ◯ a drain

3 **What does a windcatcher do?** [1]

- ◯ lifts water
- ◯ cools a building using the wind
- ◯ tells the time
- ◯ makes electricity

4 **Design a model to show one system you have learned about this year (for example, the water cycle or the digestive system). List the materials and one limitation.** [3]

My model

5 **Challenge. Explain why a city is a system. Name three parts and how they depend on each other.** [2]

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How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain what a system is, with inputs and outputs	○	○	○
explain what a model is and its limits	○	○	○
describe technologies from four civilisations	○	○	○
explain how a shadoof and a windcatcher work	○	○	○
test a model fairly	○	○	○
explain how an old technology still matters today	○	○	○